

Datasheet — finN-2e19 Smartphone Battery (LNMO 4.7 V-class)

Document: VBF-T6R1F-DSH-01 | **Case:** t6_r1_flash | **Date:** 2026-08-25 | **Values:** DFN simulation (SPM in parentheses where differing)

1. Electrical Characteristics

Parameter	Value	Conditions
Nominal capacity	6.32 Ah (6.35)	1C discharge, 25 °C, 2.5–4.7 V
Nominal energy	26.2 Wh	1C discharge
Midpoint voltage (plateau)	4.145 V (4.248)	voltage at discharge-time midpoint
Operating voltage window	2.5 – 4.7 V	CC charge cutoff 4.7 V
DC resistance	4.8 mΩ	OCV – V @10% DOD ÷ 1C
Power density	20.4 kW/kg	$V_{OC}^2/(4 \cdot DCR)/mass$
Self-heating / peak T	49.4 °C (322.5 K)	4C charge, 45 °C ambient, h=300

2. Fast-Charge Behavior (4C = 18 A, 45 °C ambient)

Parameter	Value
Charge acceptance	6.68 Ah (6.97) — real charge, 22.3 min
Max temperature (h=300)	322.5 K — ≤ 50 °C PASS
Anode min potential	–0.019 V (DFN) / –0.262 V (SPM) — plating flagged (M4 fail, see design_spec §4)
h sensitivity	326.1 K (h120) / 323.8 (h200) / 323.1 (h250) / 322.5 (h300)

3. Aging (100 cycles, 1C, isothermal)

Parameter	Value
SEI thickness end-of-test	292 nm (D_ec 2e-19) — ≤ 500 nm PASS ; bracket: 357 nm @3e-19
SEI growth regime	diffusion-limited ($L \propto \sqrt{(D_{ec} \cdot t)}$)

4. Physical Characteristics (contract caliber; electrolyte/casing excluded)

Parameter	Value
Stack thickness	217.6 μm
Footprint area	0.1027 m² (0.065 m × 1.58 m runner convention)
Mass	45.9 g
Volume	22.35 cm³
Energy density (vol)	1171.7 Wh/L
Energy density (grav)	569.9 Wh/kg
Electrolyte included	No (parameter set lacks density — contract caliber)

5. Electrode/Electrolyte System

- Cathode: LNMO spinel (LiNi■.■Mn■.■O■), 74 μm, ε 0.665, r_p = 2.0 μm, OCP ceiling 4.706 V.

- Anode: graphite, 120 μm , ϵ_{act} 0.65, r_{p} = 3.0 μm .
- Electrolyte: high-transference (t_{H}^+ 0.9), high-conductivity (σ 20 S/m, D 1e-9) additive formulation.
- SEI: LiF-rich (EC diffusivity 2e-19 m²/s).
- Cooling: vapor chamber + graphite + Al frame, effective h = 300 W/m²K (h =250 minimum verified).

6. Compliance Summary

Criterion	Value	Requirement	Verdict
ED volumetric	1171.7 Wh/L	≥ 950	PASS
Plateau voltage	4.145 V	≥ 4.1	PASS
SEI @100 cyc	292 nm	≤ 500	PASS
No plating @4C	plated	false	FAIL — documented protocol boundary (real charge retained)
T_max @4C	322.5 K	≤ 323.15	PASS

7. Caveats

- All values simulation-derived (SPMe + DFN cross-check); no hardware build.
- Electrolyte transport parameters (σ 20 S/m, t_{H}^+ 0.9) are aggressive estimates — see design_spec §6.
- The M4 fail is a system-level (LNMO cutoff) interaction, not a component defect — see design_spec §4 and DFMEA.